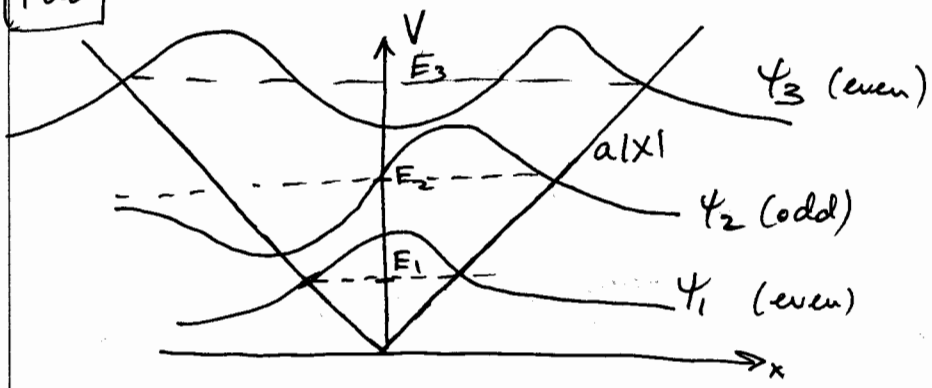
P6.8



Since $V = a|x|$ is unrestricted at $x \to \pm\infty$, the wavefunction $\psi(x)$ is exponentially suppressed there:

$$\psi(x) \underset{x\to\pm\infty}{\to} e^{-q|x|}.$$

Inside the well, it must be an oscillating function of position.

In addition, the lowest state (ψ_1) has no zeroes (nodes) inside the well; the next one (ψ_2) — has 1; the next (ψ_3) — has 2.

Finally, use the symmetry $x \to -x$ (because $V(x) = V(-x)$).

P6.11

$$V(x) = \begin{cases} 0, & 0<x<L \\ \infty, & \text{elsewhere} \end{cases}$$

we know that $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\frac{\pi n x}{L}$, $E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}$ $n=1,2,3\ldots$

$\psi_n(x)$ satisfies boundary conditions: $\psi_n(0) = 0$

$\psi_n(L) = 0$.

The ground state = lowest energy state $\Rightarrow n=1$.

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\frac{\pi x}{L}$$

$$\Delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}$$

$$\langle x\rangle = \int_0^L dx \left(\sqrt{\frac{2}{L}}\sin\frac{\pi x}{L}\right)^2 \cdot x = \begin{pmatrix} y = \frac{\pi x}{L} \\ x = \frac{L}{\pi} y \end{pmatrix} = \frac{2}{L}\int_0^\pi dy \frac{L}{\pi}\cdot\frac{L}{\pi} y \sin^2 y$$

$$= \frac{2L}{\pi^2}\int_0^\pi dy\, y \sin^2 y = \left(\sin^2 y = \frac{1-\cos 2y}{2}\right) =$$

$$= \frac{L}{\pi^2}\int_0^\pi dy\, y\,(1-\cos 2y) = \frac{L}{\pi^2}\left(\frac{y^2}{2}\Big|_0^\pi - \int_0^\pi d\left(\frac{1}{2}\sin 2y\right) y\right)$$

$$= \frac{L}{\pi^2}\left(\frac{\pi^2}{2} - \frac{y\sin 2y}{2}\Big|_0^\pi + \frac{1}{2}\underbrace{\int_0^\pi dy \sin 2y}_{\frac{1}{2}\int_0^{2\pi} dt \sin t = 0.}\right) = \frac{L}{2}$$

$\Rightarrow \langle x\rangle = \frac{L}{2}$ which is very natural.

$$\langle x^2\rangle = \int_0^L dx\, \frac{2}{L}\sin^2\left(\frac{\pi x}{L}\right) x^2 = \begin{pmatrix} y = \frac{\pi x}{L} \\ x = \frac{L}{\pi} y \end{pmatrix} =$$

$$= \frac{2}{L}\left(\frac{L}{\pi}\right)^3 \int_0^\pi dy\, y^2 \sin^2(y) = \frac{2L^2}{\pi^3}\int_0^\pi dy\, y^2 \frac{1-\cos 2y}{2}$$

$$= \frac{L^2}{\pi^3}\int_0^\pi dy\, y^2 (1-\cos 2y) = \frac{L^2}{\pi^3}\left(\frac{y^3}{3}\Big|_0^\pi - \frac{1}{8}\int_0^{2\pi} d(y\cdot 2)\,(2y)^2 \cos 2y\right)$$

$$= \frac{L^2}{\pi^3}\left(\frac{\pi^3}{3} - \frac{1}{8}\int_0^{2\pi} dt\, t^2 \cos t\right)$$

Denote $I = \int_0^{2\pi} dt\, t^2 \cos t = \int_0^{2\pi} d(\sin t)\, t^2 =$

$$= t^2 \sin t\Big|_0^{2\pi} - \int_0^{2\pi} d(t^2)\sin t = -2\int_0^{2\pi} dt \cdot t \sin t$$

$$= (-2)\int_0^{2\pi} d(-\cos t)\cdot t = (-2)\left[-t\cos t\Big|_0^{2\pi} + \int_0^{2\pi} dt \cos t\right]$$

$$= (+2)\left[2\pi + \sin t\Big|_0^{2\pi}\right] = 4\pi$$

$$\Rightarrow \quad \langle x^2 \rangle = \frac{L^2}{\pi^3}\left(\frac{\pi^3}{3} - \frac{4\pi}{8}\right) = L^2\left(\frac{1}{3} - \frac{1}{2\pi^2}\right)$$

$$\Rightarrow (\Delta x)^2 = L^2\left(\frac{1}{3} - \frac{1}{2\pi^2}\right) - \left(\frac{L}{2}\right)^2 = L^2\left(\frac{1}{3} - \frac{1}{2\pi^2} - \frac{1}{4}\right) =$$

$$= L^2\left(\frac{1}{12} - \frac{1}{2\pi^2}\right)$$

$$\Rightarrow \Delta x = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}$$

$$\langle p_x \rangle = \frac{2}{L}\int_0^L dx \, \sin\left(\frac{\pi x}{L}\right) \cdot \left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right) \sin\left(\frac{\pi x}{L}\right) =$$

$$= (-i\hbar)\,\frac{2}{L}\,\frac{\pi}{L}\int_0^L dx \, \sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right) =$$

$$= (-i\hbar)\,(2\pi)\,\frac{L}{\pi}\int_0^\pi dy \, \sin(y)\cos(y) = (-i\hbar)\cdot(2L)\cdot$$

$$\cdot \int_0^\pi d(\cos y)\cos y = (-i\hbar)\cdot L\cdot\left(\cos^2 y\right)\Big|_0^\pi = 0.$$

$$\langle p_x^2 \rangle = \frac{2}{L}\int_0^L dx \, \sin\left(\frac{\pi x}{L}\right)\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 \sin\left(\frac{\pi x}{L}\right) =$$

$$= -\frac{2\hbar^2}{L}\int_0^L dx \left(\frac{\pi}{L}\right)^2(-1)\,\sin\frac{\pi x}{L}\,\sin\frac{\pi x}{L}$$

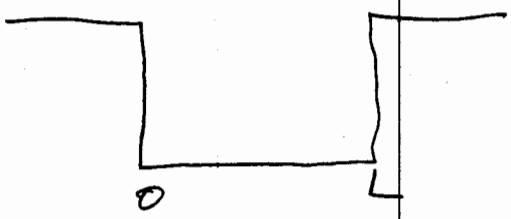
$$= \frac{2\hbar^2}{L}\left(\frac{\pi}{L}\right)^2 \frac{L}{\pi}\underbrace{\int_0^\pi dy \, \sin^2(y)}_{\pi/2} = \frac{2\hbar^2}{L}\left(\frac{\pi}{L}\right)^2\frac{L}{2} = \left(\frac{\hbar\pi}{L}\right)^2$$

$$\Rightarrow \Delta p_x = \frac{\hbar\pi}{L}$$

$$\Rightarrow \Delta x \cdot \Delta p_x = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}\;\frac{\hbar\pi}{L} = \frac{\hbar}{2}\sqrt{\frac{\pi^2}{3} - 2} = \underline{\underline{0.57\hbar}}$$

$\boxed{P6.12}$ Infinite rectangular potential well.

$$V(x) = \begin{cases} 0 & , 0 < x < L \\ \infty & , \text{elsewhere} \end{cases}$$



$$t=0 \quad \psi(x) = \begin{cases} \frac{1+i}{2}\sqrt{\frac{2}{L}}\sin\frac{\pi x}{L} + \frac{1}{\sqrt{2}}\sqrt{\frac{2}{L}}\sin\frac{2\pi x}{L} & , 0<x<L \\ 0 & , \text{elsewhere} \end{cases}$$

$$\Rightarrow \psi(x, t=0) = \frac{1+i}{2}\psi_1(x) + \frac{1}{\sqrt{2}}\psi_2(x)$$

where $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{\pi n x}{L}$ are eigenstates of the problem with eigenvalues $E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}$.

By construction, the eigenstates evolve with time as

$$\psi_n(x,t) = \psi_n(x)\cdot e^{-iE_n t/\hbar}$$

(a) $\psi(x,t) = \frac{1+i}{2}\psi_1(x,t) + \frac{1}{\sqrt{2}}\psi_2(x,t) =$

$$= \frac{1+i}{2}\psi_1(x)e^{-iE_1 t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{-iE_2 t/\hbar}$$

where $E_1 = \frac{\hbar^2\pi^2}{2mL^2}$, $E_2 = \frac{\hbar^2\pi^2(2)^2}{2mL^2}$

(b) $\langle E\rangle = |\langle\psi_1(x)|\psi(x,t)\rangle|^2 E_1 + |\langle\psi_2(x)|\psi(x,t)\rangle|^2 E_2$

$$= \left|\frac{1+i}{2}e^{-iE_1 t/\hbar}\right|^2 E_1 + \left|\frac{1}{\sqrt{2}}e^{-iE_2 t/\hbar}\right|^2 E_2 =$$

$$= \frac{1+1}{4}E_1 + \frac{1}{2}E_2 = \frac{1}{2}(E_1+E_2) = \frac{1}{2}(1+4)\frac{\hbar^2\pi^2}{2mL^2}$$

$$= \frac{5}{4}\frac{\hbar^2\pi^2}{mL^2}$$

Note that $\langle \psi_n(x) | \psi(x,t) \rangle$ = amplitude that a particle in state $\psi(x,t)$ is in the <u>eigenstate</u> $\psi_n(x)$.
The probability then is $|\langle \psi_n | \psi(x,t) \rangle|^2 = p_n$,

and any expectation value can be found as a weighted sum of expectation values in eigenstates ψ_n: $\langle A \rangle = \sum_n p_n A_n$

(c) $E = \frac{\hbar^2 \pi^2}{2mL^2} = E_1$ is the energy of state $\psi_1(x)$

$\Rightarrow$ the probability $p_1 = |\langle \psi_1 | \psi(x,t) \rangle|^2 =$

$$= \left| \frac{1+i}{2} e^{-iE_1 t/\hbar} \langle \psi_1 | \psi_1 \rangle + \frac{1}{\sqrt{2}} e^{-iE_2 t/\hbar} \langle \psi_1 | \psi_2 \rangle \right|^2$$

$$= \left| \frac{1+i}{2} \right|^2 = \frac{(1+i)(1-i)}{2 \cdot 2} = \frac{1}{2}.$$

Here I used that $\langle \psi_1 | \psi_n \rangle = \begin{cases} 1, & \text{for } n=1 \\ 0, & \text{for } n \neq 1 \end{cases}$

because the eigenstates are:
(1) normalized (2) orthogonal.

(d) $\frac{d}{dt} \langle \hat{x} \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{x}] \rangle$ (eq. 4.16)

and $[\hat{H}, \hat{x}] \neq 0$ (because $[\hat{p}_x, \hat{x}] = i\hbar$).

$\Rightarrow \frac{d}{dt} \langle \hat{x} \rangle \neq 0$ in general.

P6.14

$$\psi(x) = \begin{cases} N\, x(x-L) & 0 < x < L \\ 0 & \text{elsewhere} \end{cases}$$

Note that this $\psi(x)$ satisfies $\psi(0) = \psi(L) = 0$.

(a) normalization

$$1 = |N|^2 \int_0^L dx \left(x(x-L)\right)^2 = N^2 \int_0^L dx\, x^2 (x-L)^2 =$$

$$= (x = yL) = N^2 L^5 \int_0^1 dy\, y^2 (1-y)^2 = N^2 L^5 \int_0^1 dy\, y^2 (1 - 2y + y^2) =$$

$$= L^5 N^2 \left[\frac{1}{3} - 2\frac{1}{4} + \frac{1}{5} \right] = \frac{N^2 L^5}{30} \Rightarrow N = \sqrt{\frac{30}{L^5}}$$

(b) Probability P to get the ground state energy $E_1 = \frac{\hbar^2 \pi^2}{2mL^2}$

Clearly $P = |\langle \psi_1 | \psi \rangle|^2 = \left| \int_0^L dx \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \cdot \sqrt{\frac{30}{L^5}}\, x(x-L) \right|^2$

Calculate $I = \langle \psi_1 | \psi \rangle = \sqrt{\frac{2}{L}} \sqrt{\frac{30}{L^5}} \int_0^L dx \sin\left(\frac{\pi x}{L}\right) x(L-x)$

$$= \left(y = \frac{\pi x}{L}\right) = \sqrt{\frac{60}{L^6}}\, \frac{L}{\pi} \int_0^\pi dy \sin(y)\, \frac{L}{\pi} y \left(L - \frac{L}{\pi} y\right) =$$

$$= \sqrt{\frac{60}{L^6}} \left(\frac{L}{\pi}\right)^3 \int_0^\pi dy \sin y\, y\, (\pi - y) = \frac{\sqrt{60}}{\pi^3} \left(I_1 \cdot \pi - I_2\right)$$

where $I_1 = \int_0^\pi \sin y\, y\, dy = \int_0^\pi d(-\cos y)\, y = -y \cos y \Big|_0^\pi + \underbrace{\int_0^\pi dy \cos y}_{\sin y \big|_0^\pi = 0} = \pi.$

$$I_2 = \int_0^\pi dy\, y^2 \sin y = \int_0^\pi d(-\cos y)\, y^2 = -y^2 \cos y \Big|_0^\pi + 2\int_0^\pi dy \cdot y \cos y$$

$$= \pi^2 + 2\int_0^\pi d(\sin y)\, y = \pi^2 + 2\left[y \sin y \Big|_0^\pi - \int_0^\pi dy \sin y \right]$$

$$\Rightarrow I_2 = \pi^2 + 4$$

$$\Rightarrow I = \langle \psi_1 | \psi \rangle = \frac{\sqrt{60}}{\pi^3}\left(\pi^2 - \pi^2 - 4\right) = \frac{4\sqrt{60}}{\pi^3}$$

$$\Rightarrow P = \frac{16 \cdot 60}{\pi^6} = 0.9986$$

The probability is almost 1! $\Rightarrow \psi(x)$ is a very good approximation.

(c) $$\langle E \rangle = \int_0^L dx\, \psi(x)\, \hat{E}\, \psi(x) =$$

$$= \int_0^L dx\, \psi(x)\, \underset{\text{operator of energy}}{\hat{H}}\, \psi(x) = \int_0^L dx\, \psi(x) \left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right) \psi(x)$$

$$= \frac{30}{L^5}\left(-\frac{\hbar^2}{2m}\right) \int_0^L dx\, x(x-L)\, \frac{d^2}{dx^2}\, x(x-L) =$$

$$= -\frac{\hbar^2}{2m}\frac{30}{L^5} \int_0^L dx\, x(x-L) \cdot 2 = -\frac{\hbar^2}{2m}\frac{60}{L^5}\left(\frac{x^3}{3} - L\frac{x^2}{2}\right)\Big|_0^L =$$

$$= -\frac{\hbar^2}{2m}\frac{60}{L^5}\left(\frac{L^3}{3} - \frac{L^3}{2}\right) = -\frac{\hbar^2}{2m}\frac{60}{L^5}\left(-\frac{L^3}{6}\right) =$$

$$= \frac{\hbar^2 \cdot 10}{2mL^2} = \frac{\hbar^2 \cdot \pi^2}{2mL^2} \cdot \frac{10}{\pi^2} = 1.013 \cdot E_1$$

$$\frac{\langle E \rangle - E_1}{E_1} = 0.013$$ very close to the correct ground state energy

Important Problem $\boxed{P\,6.16}$ $$\frac{2m}{\hbar^2} V(x) = -\frac{\lambda}{6}\delta(x)$$

(a) $$\Rightarrow \text{S.E.} \quad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

$$\frac{d^2\psi}{dx^2} - \frac{2m}{\hbar^2}V(x)\psi = -\frac{2m}{\hbar^2}E\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{\lambda}{6}\delta(x)\psi = -\frac{2mE}{\hbar^2}\psi \quad (1)$$

Integrate both sides of (1) in the interval $(-\varepsilon, \varepsilon)$

$$\int_{-\varepsilon}^{\varepsilon} dx \, \frac{d^2\psi}{dx^2} = \frac{d\psi(\varepsilon)}{dx} - \frac{d\psi(-\varepsilon)}{dx}$$

$$\int_{-\varepsilon}^{\varepsilon} dx \, \frac{\lambda}{b} \delta(x) \psi(x) = \frac{\lambda}{b} \psi(0) \; ;$$

$$\int_{-\varepsilon}^{\varepsilon} dx \, \frac{2mE}{\hbar^2} \psi(x) \xrightarrow[\varepsilon \to 0]{} 0$$

$$\Rightarrow \quad \frac{d\psi(+0)}{dx} - \frac{d\psi(-0)}{dx} = -\frac{\lambda}{b} \psi(0) \quad (2)$$ that is, derivative jumps at $x=0$.

(b) let $E<0$; denote $\varkappa^2 = -\frac{2mE}{\hbar^2} > 0$.

For $\underline{x>0}$ $\quad \frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi = \varkappa^2 \psi$

$$\Rightarrow \psi(x) = A e^{-\varkappa x}$$

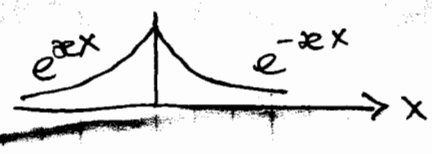
For $\underline{x<0}$ $\quad \frac{d^2\psi}{dx^2} = \varkappa^2 \psi \Rightarrow \psi(x) = B e^{\varkappa x}$

Wave function is continuous: $A = B$

Derivatives is discontinuous: using (2), we derive

$$-\varkappa A - (\varkappa \overset{A}{\overset{\shortparallel}{B}}) = -\frac{\lambda}{b} A$$

$$\Rightarrow -2\varkappa A = -\frac{\lambda}{b} A \quad \Rightarrow \boxed{\varkappa = \frac{\lambda}{2b}}$$

$$\Rightarrow E = -\frac{\hbar^2}{2m}\varkappa^2 = -\frac{\hbar^2}{2m}\left(\frac{\lambda}{2b}\right)^2 = -\frac{\hbar^2 \lambda^2}{8mb^2}$$

$\Rightarrow$ there is <u>one</u> <u>bound</u> state